

AS/NZS IEC 62471:2011

IEC 62471, Ed.1.0 (2006)

Australian/New Zealand Standard™ Photobiological safety of lamps and lamp systems

PREFACE

This Standard was prepared by the Joint Standards Australia/Standards New Zealand Committee

SF-019, Personal Protection Against Laser Radiation.

The objectives of this Standard are as follows:

(a) To protect people from optical radiation in the wavelength range 200 nm to 3000 nm by introducing a system of classification for all electrically powered incoherent broadband sources of optical radiation, including light emitting diodes (LEDs) but excluding lasers, according to their degree of optical radiation hazard.

(b) To provide a scheme for evaluation and control of photobiological hazards through specification of exposure limits and reference measurement technique.

(c) To ensure adequate warnings are provided to individuals of hazards associated with accessible incoherent broadband sources of optical radiation through the use of labels and instructions

(refer to next paragraph).

This is the first Standard adopted by Australia/New Zealand relating specifically to the photobiological safety of lamps and lamp systems and is normative. However, additional informative guidance may be found in IEC/TR 62471-2, Edition 1.0 (2009)*, Photobiological safety of lamps and lamp systems—Part 2: Guidance on manufacturing requirements relating to non-laser optical radiation safety. The aforementioned IEC technical report provides in the context of classifications defined in AS/NZS IEC 62471, guidance on: requirements for optical radiation safety assessment; allocation of safety measures; and labelling of products. This Standard is identical with, and has been reproduced from, IEC 62471, Ed.1.0 (2006), Photobiological safety of lamps and lamp systems. As this Standard is reproduced from an International Standard, the following applies:

(a) Its number appears on the cover and title page while the International Standard number appears only on the cover.

(b) In the source text 'This International Standard' should read 'This Australian/New Zealand

Standard'.

(c) A full point substitutes for a comma when referring to a decimal marker.

None of the normative references in the source document have been adopted as Australian or Australian/New Zealand Standards.

The term 'informative' has been used in this Standard to define the application of the annex to which it applies. An 'informative' annex is only for information and guidance.

Photobiological safety of lamps and lamp systems

PHOTOBIOLOGICAL SAFETY OF LAMPS

AND LAMP SYSTEMS

1. SCOPE

This International Standard gives guidance for evaluating the photobiological safety of lamps and lamp systems including luminaires. Specifically it specifies the exposure limits, reference measurement technique and classification scheme for the evaluation and control of photobiological hazards from all electrically powered incoherent broadband sources of optical radiation, including LEDs but excluding lasers, in the wavelength range from 200 nm through 3000 nm.

2. NORMATIVE REFERENCES

The following referenced documents are indispensable for the application of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies. CIE 17.4-1987 *International lighting vocabulary* (ILV) – Joint publication IEC/CIE CIE 53-1982 *Methods of characterizing the performance of radiometers and photometers* CIE 63-1984 *The spectroradiometric measurement of light sources* CIE 105-1993 *Spectroradiometry of pulsed optical radiation sources*

ISO *Guide to the expression of uncertainty in measurement*, ISO, Geneva, 1995.

3. DEFINITIONS, SYMBOLS AND ABBREVIATIONS

For the purposes of this standard, the following definitions, symbols and abbreviations apply.

3.1 actinic dose (see ILV 845-06-23)

Quantity obtained by weighting spectrally the dose according to the actinic action spectrum value at the corresponding wavelength.

Unit: J·m⁻²

Note: This definition implies that an action spectrum is adopted for the actinic effect considered, and that its maximum value is generally normalized to 1. When giving a quantitative amount, it is essential to specify which quantity dose or actinic dose is meant, as the unit is the same.

3.2 angular subtense (α)

Visual angle subtended by the apparent source at the eye of an observer or at the point of measurement. In this standard subtended angles are denoted by the full included angle, not the half angle.

Unit: radian

Note: The angular subtense α will generally be modified by incorporation of lenses and mirrors as projector optics, i.e. the angular subtense of the apparent source will differ from the angular subtense of the physical source.

3.3 aperture, aperture stop

Opening that defines the area over which average optical emission is measured. For spectral irradiance measurements this opening is usually the entrance of a small sphere placed in front of the radiometer/spectroradiometer entrance slit.

3.6 erythema (see ILV 845-06-15)

Reddening of the skin; as used in this standard the reddening of the skin resulting from inflammatory

effects from solar radiation or artificial optical radiation.

Note: The degree of delayed erythema is used as a guide to dosages applied in ultraviolet therapy.

3.7 exposure distance

Nearest point of human exposure consistent with the application of the lamp or lamp system. For lamps radiating in all directions the distance is measured from the centre of the filament or arc source. For reflector-type lamps the distance is measured from the

outside edge of the lens or the plane defining the end of the reflector in a lens free reflector.

Unit: m

3.8 exposure limit (EL)

Level of exposure to the eye or skin that is not expected to result in adverse biological effects.

3.10 field of view

Solid angle as "seen" by the detector (acceptance angle), such as the radiometer/ spectroradiometer, out of which the detector receives radiation.

Unit: sr

Note 1: The field of view should not be confused with the angular subtense of the apparent source α .

Note 2: A plane angle is sometimes used to describe a circular symmetric solid angle field of view.

3.12 hazard distance

See skin hazard distance or ocular hazard distance.

3.13 illuminance (at a point of a surface) (E_v) (see ILV 845-01-38)

Quotient of the luminous flux $d\Phi$

v incident on an element of the surface containing the point, by the area dA of that element.

$$E_v = \frac{d\Phi_v}{dA} \quad (3.1)$$

Unit: lx

Unit: lx

3.14 infrared radiation (IR) (see ILV 845-01-04)

Optical radiation for which the wavelengths are longer than those for visible radiation.

Note: For infrared radiation, the range between 780 nm and 106 nm is commonly subdivided into: IRA (780 nm to 1400 nm), IR-B (1400 nm to 3000 nm), and IR-C (3000 nm to 106 nm).

Infrared radiation is often evaluated in terms of the spectral total radiation per unit area (irradiance) incident upon a surface. Examples of applications of infrared radiation are industrial heating, drying, baking, and photo-reproduction. Some applications, such as

infrared viewing systems, involve detectors sensitive to a restricted range of wavelengths. In these cases, the spectral characteristics of the source and detector are of importance.

3.15 intended use

Use of a product, process or service in accordance with specifications, instructions and information provided by the supplier.

3.16 irradiance (at a point of the surface) (see ILV 845-01-37)

Quotient of the radiant flux $d\Phi$ incident on an element of a surface containing the point, by the area dA

of that element, i.e.,

$$E = \frac{d\Phi}{dA} \quad (3.2)$$

Unit: $\text{W}\cdot\text{m}^{-2}$

3.17 lamp (see ILV 845-07-03)

Source made to produce optical radiation, usually visible. Note: The term "lamp" is sometimes used for certain types of luminaires. These types of luminaires consist of a lamp with shade, reflector, enclosing globe, housing, or other accessories. As used in this standard, the term means an electrically powered source, other than a laser, that produces radiation in the visible region of the electromagnetic spectrum. Devices that generate light and have integral components for optical control, such as lenses or reflectors, also are considered lamps. Examples include a lensed LED, lens-end lamp, and reflector types, that consist of a source within a parabolic or elliptical reflector assembly, normally including a lens cover.

3.18 lamp system

Any manufactured product or assemblage of components that incorporates or is intended to incorporate a lamp.

3.21 light

See visible radiation.

3.22 light emitting diode (LED) (see ILV 845-04-40)

Solid state device embodying a p-n junction emitting optical radiation without gain when excited by an electric current.

3.23 lumen (see ILV 845-01-51)

SI unit of luminous flux: Luminous flux emitted in a unit solid angle (steradian) by a uniform point source having a luminous intensity of 1 candela, or equivalently, the luminous flux of a beam of monochromatic radiation whose frequency is $540 \cdot 10^{12}$ hertz and whose radiant flux is 1/683 watt.

3.24 luminaire (see ILV 845-10-01)

Apparatus which distributes, filters or transforms the light emitted from one or more lamps and which includes, except the lamps themselves, all the parts necessary for fixing and protecting the lamps and, where necessary, circuit auxiliaries together with the means for connecting them to the electric supply. The words "luminaire" and "lamp system" are often assumed to be synonymous. For the purposes of this standard, the word "luminaire" is restricted to apparatus used for distributing light in general lighting, while "lamp system" implies use of lamps in other than general lighting applications.

3.25 luminance (in a given direction, at a given point of a real or imaginary surface) (**L_v**) (see ILV 845-01-35)

Quantity defined by the formula

$$L_v = \frac{d\Phi_v}{dA \cdot \cos \theta \cdot d\Omega} \quad (3.3)$$

where $d\Phi_v$ is the luminous flux transmitted by an elementary beam passing through the given point and propagating in the solid angle $d\Omega$ containing the given direction; dA is the area of a section of that beam containing the given point; θ is the angle between the normal to that section and the direction of the beam.

Unit: $\text{cd} \cdot \text{m}^{-2}$

3.26 lux (see ILV 845-01-52)

SI unit of illuminance: Illuminance produced on a surface of area 1 square metre by a luminous flux of

1 lumen uniformly distributed over that surface.

3.27 ocular hazard distance

Distance from a source within which the radiance or irradiance for a given exposure duration exceeds the applicable exposure limit.

Unit: m

3.28 optical radiation (see ILV 845-01-02)

Electromagnetic radiation at wavelengths between the region of transition to X-rays (wavelength approximately 1 nm) and the region of transition to radio waves (wavelength approximately 106 nm). Ultraviolet radiation in the wavelength range below

180 nm (vacuum UV) is strongly absorbed by the oxygen in air. For the purpose of this standard the wavelength band of optical radiation is limited to wavelengths greater than 200 nm. Further, the eye transmits optical radiation to the retina between 380 and 1400 nm. Thus this wavelength range requires special consideration in determining the photobiological safety of the retina.

3.29 photokeratoconjunctivitis

Inflammatory response of the cornea and conjunctiva following exposure to ultraviolet (UV) radiation.

Wavelengths shorter than 320 nm are most effective in causing this condition. The peak of the action spectrum is approximately at 270 nm.

Note: Different action spectra have been published for photokeratitis and photoconjunctivitis (CIE 106/2 and CIE 106/3–1993); however, the latest studies support the use of a single action spectrum for both ocular effects (CIE 106/1–1993).

3.30 pulsed lamp

Lamp that delivers its energy in the form of a single pulse or a train of pulses where each pulse is assumed to have a duration of less than 0,25 s. A lamp with a continuous train of pulses or modulated radiant energy where the peak radiated power is at least ten times the average radiated power.

Note 1: The **duration** of a lamp pulse is the time interval between the half-power points on the leading and the trailing edges of the pulse.

Note 2: In this standard, General lighting service lamps are defined to be Continuous wave lamps (see clause 3.5). Examples of pulsed lamps include photoflash lamps, flash lamps in photocopy machines, pulse-modulated LEDs, and strobe lights.

3.31 radiance (in a given direction at a given point of a real or imaginary surface) (**L**) (see ILV 845-01-34)

Quantity defined by the formula,

where $d\Phi$ is the radiant power (flux) transmitted by an elementary beam passing through the given

point and propagating in the solid angle $d\Omega$ containing the given direction; dA is the area of a section of

that beam containing the given point; θ is the angle between the normal to that section and the

direction of the beam.

$$L = \frac{d\Phi}{dA \cdot \cos \theta \cdot d\Omega} \quad (3.4)$$

Unit: W·m⁻²·sr⁻¹

The same definition holds for the time-integrated radiance L_i if, in the equation for L , the radiant power $d\Phi$ is replaced by the radiant energy dQ .

3.32 radiant energy (see ILV 845-01-27)

Time integral of the radiant power, Φ over a given duration, Δt .

3.32 radiant energy (see ILV 845-01-27)

Time integral of the radiant power, Φ over a given duration, Δt .

$$Q = \int_0^t \Phi \cdot dt \quad (3.5)$$

Unit: J

Unit: J

3.33 radiant exposure (at a point of a surface, for a given duration) (see ILV 845-01-42)

Quotient of the radiant energy, dQ , incident on an element of the surface containing the point over the given duration, by the area dA of that element.

$$H = \frac{dQ}{dA} \quad (3.6a)$$

Unit: J·m⁻²

Equivalently the radiant exposure is defined as the integral of the irradiance, E , at a given point over a given duration, Δt .

$$H = \int_{\Delta t} E \cdot dt \quad (3.6b)$$

3.34 radiant power (Φ) (see ILV 845-01-24)

Power emitted, transmitted or received in the form of radiation. Radiant power is often called *radiant flux*.

Unit: watt (W)

3.38 skin hazard distance

Distance at which the irradiance exceeds the applicable exposure limit for 8 hours exposure.

Unit: m

3.39 spectral distribution (see ILV 845-01-17)

Quotient of the radiant, luminous or photon quantity $dX(\lambda)$ contained in an elementary range $d\lambda$ of wavelength at the wavelength λ , by that range.

$$X_\lambda = \frac{dX(\lambda)}{d\lambda} \quad (3.7)$$

Unit: $[X] \cdot \text{nm}^{-1}$

Note: The term spectral distribution is to be preferred when dealing with the function $X_\lambda(\lambda)$ over a wide range of wavelengths, not at a particular wavelength.

3.40 spectral irradiance

Quotient of the radiant power $d\Phi(\lambda)$ in a wavelength interval $d\lambda$, incident on an element of a surface, by the area dA of that element and by the wavelength interval $d\lambda$.

$$E_\lambda = \frac{d\Phi(\lambda)}{dA \cdot d\lambda} \quad (3.8)$$

Unit: $\text{W} \cdot \text{m}^{-2} \cdot \text{nm}^{-1}$

3.41 spectral radiance (for a wavelength interval $d\lambda$, in a given direction at a given point) (L_λ)

Ratio of the radiant power $d\Phi(\lambda)$ passing through that point and propagating within the solid angle $d\Omega$ in the given direction, to the product of the wavelength interval $d\lambda$ and the area of a section of that beam on a plane perpendicular to this direction ($\cos \theta \, dA$) containing the given point and to the solid angle $d\Omega$.

$$L_\lambda = \frac{d\Phi(\lambda)}{dA \cdot \cos \theta \cdot d\Omega \cdot d\lambda} \quad (3.9)$$

Unit: $\text{W} \cdot \text{m}^{-2} \cdot \text{nm}^{-1} \cdot \text{sr}^{-1}$

3.42 steradian (see ILV 845-01-20)

SI unit of solid angle. A solid angle that, having its vertex at the centre of a sphere, cuts off an area of the surface of the sphere equal to that of a square with sides of length equal to the radius of the sphere.

3.45 visual angle

Angle subtended by an object or detail at the point of observation is considered to be the visual angle. The SI unit for the angle is the radian although it may also be measured in milliradians, degrees, or minutes of arc.

4. EXPOSURE LIMITS (EL'S)

4.1 General

Individuals in the vicinity of lamps and lamp systems shall not be exposed to levels exceeding the limits developed in the following sections. The exposure limit (EL) values are taken from various ICNIRP guidelines which, in turn, are based on the best available information from experimental studies (see Annex A for summary of publications).

The exposure limits represent conditions under which it is believed that nearly all individuals in the general population may be repeatedly exposed without adverse health effects. However, they do not apply to abnormally photosensitive individuals or to individuals concomitantly exposed to photosensitizing agents, which makes individuals much more susceptible to adverse health effects from optical radiation. Such individuals, in general, are more susceptible to adverse health effects from optical radiation than individuals who are not abnormally photosensitive or concomitantly exposed to photosensitizing agents. The susceptibility of photosensitive individuals varies greatly, and it is not possible to set exposure limits for this portion of the population.

The exposure limits in this standard apply to continuous sources where the exposure duration is not less than 0,01 ms and not more than any 8-hour period, and should be used as guides in the control of exposure. The values should not be regarded as precisely defined lines between safe and unsafe levels.

The limits for exposure to broad-band visible and IR-A radiation for the eye require knowledge of the spectral radiance of the source, L_λ , and total irradiance, E , as measured at the position(s) of the eye of the exposed person. Such detailed spectral data of a light source are generally required only if the luminance of the source exceeds 104 cd·m⁻². At a luminance less than this value, the exposure limits are expected to be not exceeded. The exposure limits are given in clause 4.3.

4.3 Hazard exposure limits

4.3.8 Thermal hazard exposure limit for the skin

Visible and infrared radiant exposure (380 nm to 3000 nm) of the skin shall be limited to:

$$E_H \cdot t = \sum_{380}^{3000} \sum_t E_\lambda(\lambda, t) \cdot \Delta t \cdot \Delta \lambda \leq 20000 \cdot t^{0,25} \quad \text{J} \cdot \text{m}^{-2} \quad (t \leq 10 \text{ s}) \quad (4.12)$$

where:

$E_\lambda(\lambda, t)$ is the spectral irradiance in W·m⁻²·nm⁻¹,

$\Delta \lambda$ is the bandwidth in nm,

t is the exposure time in seconds.

Note: This exposure limit is based on skin injury due to a rise in tissue temperature and applies only to small area irradiation. Exposure limits for periods greater than 10 s are not provided. Severe pain occurs below the skin temperature required for skin injury, and an individual's exposure normally will be limited for comfort. Large area irradiation and heat stress are not evaluated since this involves consideration of heat exchange between the individual and the environment, physical activity, and various other factors, which cannot be applied in a product safety standard, but must be evaluated by environmental heat-stress criteria.

5. MEASUREMENT OF LAMPS AND LAMP SYSTEMS

The measurement of optical radiation for the purpose of computing photobiological radiation values poses significant challenges for the radiometrist. Typical photobiological action spectra such as $SUV(\lambda)$ have rapidly changing values with slight change in wavelength. Furthermore, transmission of radiation from lamp sources with glass envelopes have rapidly increasing output with increasing wavelength in the region where $SUV(\lambda)$ is rapidly decreasing. Hence issues of accuracy of the weighted results must be thoroughly considered. While irradiance measurements are routinely performed, radiance measurements are not routine and often difficult to make, especially for the photobiological hazards, as they involve a field of view that changes depending on the hazard evaluation. For these reasons it was thought necessary to include a rather lengthy discussion on the conditions and procedures needed to make emission measurements that will be used to assign risk group classification of various lamps and lamp systems. It should be noted that the measurement procedures described in this standard are designed to account for biophysical phenomena. Specifically, they can involve averaging over apertures or field-of-views which would be considered inappropriate for general radiometric measurements. However, hazards might be overestimated if non-averaged measurement values were to be compared with the respective exposure limits.

To better provide a comparison of the various exposure limits, developed in clause 4.3, including the effects of the field-of-view, both a tabular and graphical summary are shown at the end of this clause. Thus Figure 5.4 and Table 5.4 summarizes the maximum values for each of the irradiance based hazard exposure quantities as a function of exposure time, while Figure 5.5 and Table 5.5 summarizes the maximum radiance based (retinal) hazard exposure quantities, also as a function of exposure time.

Note: The upper wavelength range for evaluation of any hazard is given as 3000 nm in clause 1.

Spectral irradiance or radiance measurements using a monochromator are often difficult to make in the IR, particularly between 2500 and 3000 nm due to a lack of signal response and difficulty in obtaining calibrated sources. However, no weighting function is defined at wavelengths greater than 1400 nm. Thus broadband measurements for wavelengths between 1400 and 3000 nm are suitable in evaluating IR hazard conditions for the eye and skin in this region.

5.1 Measurement conditions

Measurement conditions shall be reported as part of the evaluation against the exposure limits and the assignment of risk classification.

5.1.1 Lamp ageing (seasoning)

To maintain stable output during the measurement process and provide reproducible results, lamps shall be seasoned for an appropriate period of time. During the initial period of operation a lamp output characteristic will change as its components come to near equilibrium. If measurements are taken of an unseasoned lamp, the variations within the measurement period and between measurements could be significant. As the output of a lamp generally decreases over life, the seasoning time should be short to result in conservative hazard evaluations. Seasoning of lamps shall be done as stated in the appropriate IEC lamp standard. Note: The seasoning time for discharge lamps, e.g., fluorescent or High Intensity Discharge (HID) types, is typically 100 h, for tungsten lamps it is on the order of one percent of rated lamp life. However, these seasoning criteria may differ for special applications, e.g. for solaria lamps.

5.1.2 Test environment

The accurate measurement of light sources requires a controlled environment. The operation of sources and measurement equipment is impacted by environmental factors. Additionally the formation of ozone in the measurement path may compromise accuracy and may present a safety hazard. For specific test conditions, see the appropriate IEC lamp standard or in the absence of such standards, the appropriate national standards or manufacturer's recommendations.

The ambient temperature will significantly influence the output of certain light sources; e.g., fluorescent lamps. The ambient temperature in which measurements are taken shall be maintained in accordance with the appropriate IEC lamp standard.

The characteristics of some light sources are also significantly affected by draughts. Air movement over the surface of test lamps, other than that caused by natural convection due to the lamp itself, should be reduced as much as possible consistent with safety considerations (ozone production). When the system under test provides interlocks that maintain circulation, measurements shall be performed with circulation.

5.1.3 Extraneous radiation

Careful checks should be made to ensure that extraneous sources of radiation and reflections do not add significantly to the measurement results. Often baffles are used to reduce extraneous radiation. Note that visually black surfaces can be reflective to UV and IR radiation. In addition radiation from hot baffles must be considered in infrared measurements due to the large input angle subtended by baffles.

5.1.4 Lamp operation

Operation of the test lamp shall be provided in accordance with the appropriate IEC lamp standard. If no standard for the lamp type exists, the lamp manufacturer's recommendation for operation should be used.

5.1.5 Lamp system operation

The power source for operation of the test lamp shall be provided in accordance with the appropriate IEC standard. If no standard for the control gear exists, the lamp manufacturer's recommendation for operation should be used.

5.2 Measurement procedure

5.2.1 Irradiance measurements

The description given applies both to broadband and spectral irradiance measurements. An ideal instrument to measure irradiance involves a plane circular area detector of diameter D , sufficient to achieve the desired signal-to-noise ratio and, that:

- accepts radiation within a right circular cone whose centerline is normal to the plane of the detector area,
- has an angular spatial response varying as the cosine of the angle from the normal to the detector area,
- has a spectral response that is constant with position within a specified wavelength band from λ_1 to λ_2 .

In this standard the minimum input aperture diameter shall be 7 mm with a maximum input aperture diameter of 50 mm. A plane circular aperture of 25 mm diameter is common on small integrating spheres, recommended above as the input for monochromators. The 25 mm diameter aperture is recommended for sources with spatially uniform optical radiation patterns. For sources that do not produce a spatially uniform irradiance, e.g., narrow beam reflector lamps, the peak irradiance (intensity) may be significantly higher than that obtained by measurement using the under-filled 25 mm diameter aperture. In such cases the detector aperture should be limited to a 7 mm diameter aperture.

Figure 5.1 schematically shows the main concepts involved in making irradiance or spectral irradiance measurements, including, if needed, the aperture to limit the field of view, of half angle, A , at some distance from the receptor that is large with respect to the detector diameter.

The measurement shall be made in that position of the beam giving the maximum reading. The instrument shall be calibrated to read in absolute incident radiant power per unit receiving area.

Note 1: From a practical point changing the input aperture requires significant extra work in recalibration of the radiometer or spectroradiometer. If the change of irradiance with distance is known, one method of achieving the lower aperture requirement is to move the detector aperture (assume 25 mm diameter is used) away from the source to a measurement distance where the 7 mm aperture cone at a distance of 200 mm now fills the 25 mm aperture, i.e., a distance about 3,5 times the standard evaluation distance.

Note 2: The measured irradiance should not be averaged over a smaller aperture than specified, as this might result in an overestimation of the hazard. The minimum size of the averaging aperture is related to physiological and behavioural factors that result in the averaging of the incident radiation over a certain surface area.

Note 3: For a number of reasons, including the physiology of the eye, all the exposure levels for ultraviolet radiation discussed in clause 4.3.1 and 4.3.2 apply to sources that subtend an angle less than 80 degrees (1,4 radian), i.e., sources within 40 degrees of the normal to the irradiance area. Thus emission from sources that subtend a greater angle need to be measured only over a full angle of 80 degrees.

Irradiance measurements apply to the following hazards described in clause 4.3 of this standard.

- 315 nm – 400 nm ocular exposure limit, *EUVA*
- IR hazard exposure limit, *EIR*
- Skin – thermal hazard exposure limit, *EH*

Spectral irradiance measurements apply to the following hazards also described in clause 4.3 of this standard.

- 200 – 400 nm skin and ocular exposure limit, *ES*
- Retinal blue light hazard exposure limit – small source, *EB*

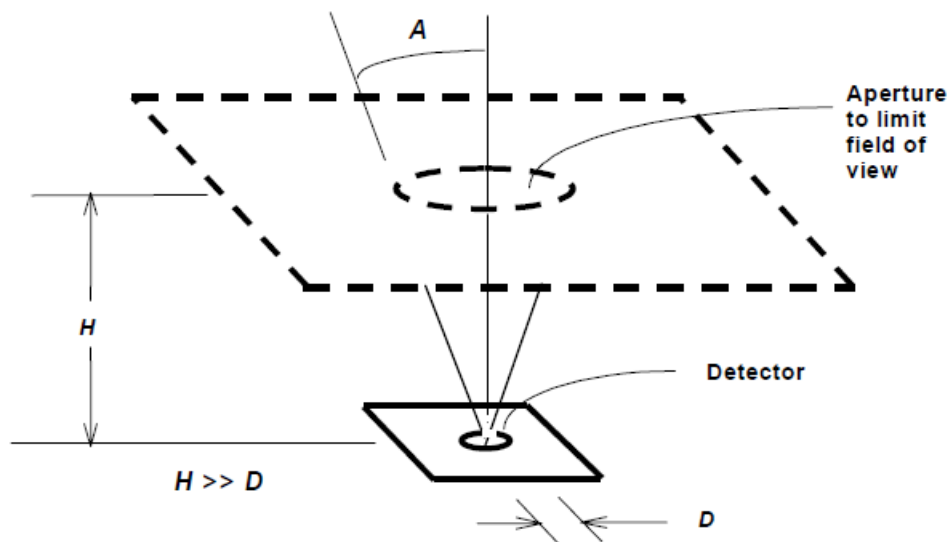


Figure 5.1 Schematic - Irradiance measurements.

5.2.2 Radiance measurements

5.2.2.1 Standard method

The description given applies both to broadband and spectral radiance measurements. Radiance measurements are performed with an optical system that (see Figure 5.2):

- images the radiant source onto a detector,
- has a circular field stop to establish the specified angular extent of the averaging field of view α_{eff} ,
- has a circular entrance pupil (aperture stop) which acts as an averaging aperture in the sense of irradiance measurements and fulfils the same requirements as stated in clause 5.2.1. For small angles the relationship between the detector diameter and the focal distance of the imaging device is seen to be $d = \alpha_{\text{eff}} \cdot H$.

As with irradiance measurements, the minimum aperture stop diameter D , as shown in Figure 5.2, corresponds to a 7 mm pupil diameter for pulsed sources and is a biophysically recognised averaging aperture for cw sources where the pupil could be smaller but eye and head movements allow for this aperture averaging. As with irradiance measurements, the aperture stop can exceed 7 mm if the incident irradiance profile is sufficiently uniform.

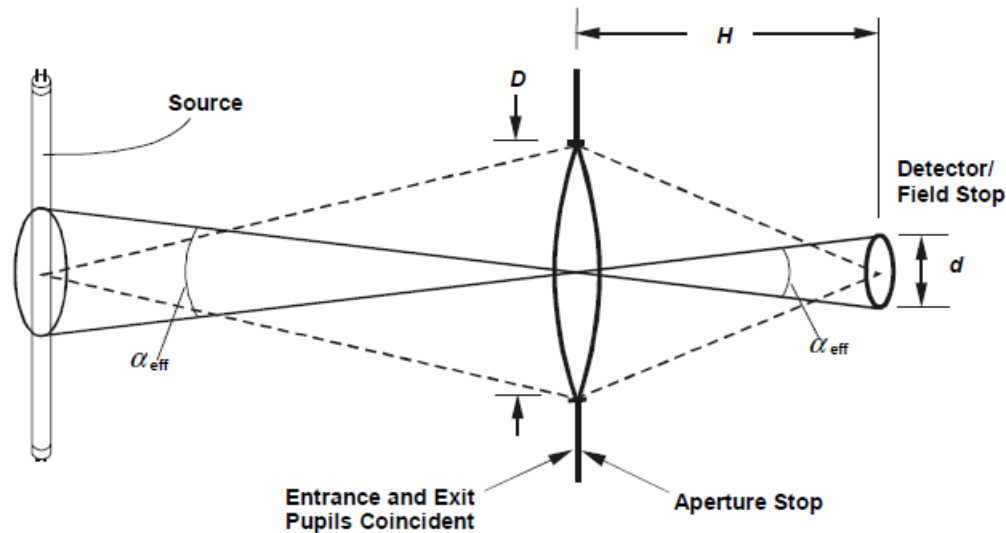


Figure 5.2 Example of an imaging device for radiance measurements.

The instrument shall be calibrated to read in absolute incident radiant power per unit receiving area and per unit solid angle of acceptance averaged over the field of view (FOV) of the instrument.

Note: The measured radiance should not be averaged over a smaller field of view than specified, as this might result in an overestimation of the hazard. The size of the averaging field of view is related to the extent of eye movements that distribute the radiant power of the source image over a larger area on the retina. The size of the averaging field of view α_{eff} is independent of the source size α . For sources that subtend an angle α smaller than the specified field of view α_{eff} , the averaged radiance value will be smaller than the actual physical radiance of the source; however this biologically effective value is the appropriate value to be compared to the exposure limit.

Spectral radiance measurements apply to the following hazards also described in clause 4.3 of this standard.

- Retinal blue light hazard exposure limit, *LB*
- Retinal thermal exposure limit, *LR*
- Retinal thermal exposure limit – weak visual stimulus, *LIR*.

5.2.4 Pulse width measurement for pulsed sources

The determination of Δt , the nominal pulse duration of a source, requires the determination of the time during which the emission is $> 50\%$ of its peak value. Common methods, e.g., using a photocell with an oscilloscope, should be applied only after verifying that the spectral uniformity is sufficient to warrant the use of visible

radiation as an analog for the UV or IR radiation. Changes in spectra during a pulse can lead to different pulse widths in different regions of the spectra.

5.3 Analysis methods

5.3.1 Weighting curve interpolations

The weighting curves defined in Table 4.1 are normally not sufficiently resolved to perform the weighted source emission calculations. The functions are reasonably linear in any local region on semi-log coordinates. Therefore to standardize interpolated values, use linear interpolation on the log of given values to obtain intermediate points at the wavelength intervals desired, e.g., one nanometer interval recommended. Anti-logarithm of the interpolated numbers result in the values needed for the interpolated weighting factors.

5.3.2 Calculations

The calculation of source hazard values shall be performed by weighting the spectral scan by the appropriate function and calculating the total weighted energy. To provide a repeatable method this standard suggests interpolation or summing to one nanometer (1 nm) for the spectra below 400 nm. Weighting and summations are then performed at this 1 nm resolution. Above 400 nm, a step size of 5 nm is recommended.

5.3.3 Measurement uncertainty

The quality of all measurement results must be quantified by an analysis of the uncertainty. All calculated results must be paired with uncertainty values that conform to the guidance in the normative references. The uncertainty of each result will be reported as the expanded uncertainty, which is calculated from the combined standard uncertainty, uc , by use of a coverage factor, $k=2$, as defined in the ISO guide listed in Clause 2. The values of uncertainty should be propagated from the calibration uncertainties, through the calculations and include all sources as described in Annex C.

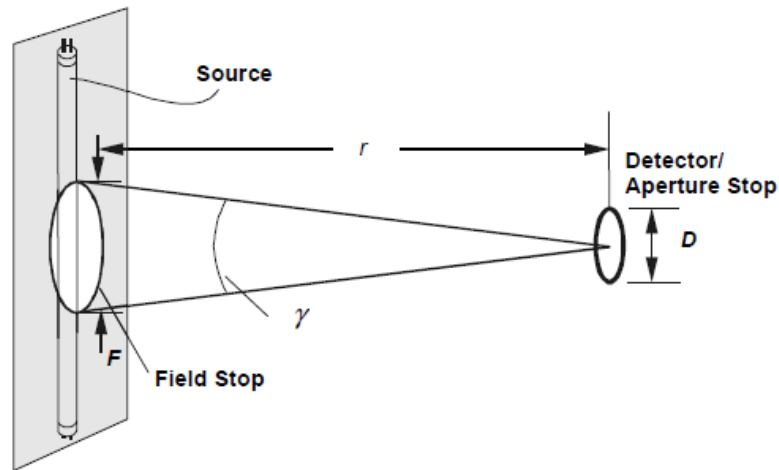


Figure 5.3 Alternative radiance technique.

Table 5.4 Summary of the ELs for the surface of the skin or cornea (irradiance based values)

Hazard Name	Relevant equation	Wavelength range nm	Exposure duration sec	Limiting aperture rad (deg)	EL in terms of constant irradiance $W \cdot m^{-2}$
Actinic UV skin & eye	$E_s = \sum E_\lambda \cdot S(\lambda) \cdot \Delta\lambda$	200 – 400	< 30000	1,4 (80)	30/t
Eye UV-A	$E_{UVA} = \sum E_\lambda \cdot \Delta\lambda$	315 – 400	≤ 1000 > 1000	1,4 (80)	10000/t 10
Blue-light small source	$E_B = \sum E_\lambda \cdot B(\lambda) \cdot \Delta\lambda$	300 – 700	≤ 100 > 100	< 0,011	100/t 1,0
Eye IR	$E_{IR} = \sum E_\lambda \cdot \Delta\lambda$	780 – 3000	≤ 1000 > 1000	1,4 (80)	18000/t ^{0,75} 100
Skin thermal	$E_H = \sum E_\lambda \cdot \Delta\lambda$	380 – 3000	< 10	2π sr	20000/t ^{0,75}

Table 5.5 Summary of the ELs for the retina (radiance based values)

Hazard Name	Relevant equation	Wavelength range nm	Exposure duration sec	Field of view radians	EL in terms of constant radiance $W \cdot m^{-2} \cdot sr^{-1}$
Blue light	$L_B = \sum L_\lambda \cdot B(\lambda) \cdot \Delta\lambda$	300 – 700	0,25 – 10 10-100 100-10000 ≥ 10000	$0,011 \cdot \sqrt{(t/10)}$ 0,011 $0,0011 \cdot \sqrt{t}$ 0,1	$10^6/t$ $10^6/t$ $10^6/t$ 100
Retinal thermal	$L_R = \sum L_\lambda \cdot R(\lambda) \cdot \Delta\lambda$	380 – 1400	< 0,25 0,25 – 10	0,0017 $0,011 \cdot \sqrt{(t/10)}$	$50000/(\alpha \cdot t^{0,25})$ $50000/(\alpha \cdot t^{0,25})$
Retinal thermal (weak visual stimulus)	$L_{IR} = \sum L_\lambda \cdot R(\lambda) \cdot \Delta\lambda$	780 – 1400	> 10	0,011	6000/α

6.2 Pulsed lamps

Pulsed lamp criteria shall apply to a single pulse and to any group of pulses within 0,25 second. A pulsed lamp shall be evaluated at the highest nominal energy loading as specified by the manufacturer.

The relevant weighted radiant exposure, (H or $E \cdot t$), or time-integrated weighted radiance dose, ($L \cdot t$), for each pulse shall be obtained by integration of the weighted irradiance or radiance emitted from the source over the full pulse width, with the integration time limited to a maximum of 0,25 s. The weighted radiant exposure or weighted radiance dose calculated shall be compared to the exposure limits (ELs) given in clause 4.3 for each of the photobiological hazards evaluated.

Note: The weighted radiance values obtained shall be averaged over a right circular cone field of view of 0,0017 radian included angle as discussed in clause 4.2.2.

The risk group determination of the lamp being tested shall be made as follows:

- A lamp that exceeds the exposure limit shall be classified as belonging to Risk Group 3 (High-Risk).
- For single pulsed lamps, a lamp whose weighted radiant exposure or weighted radiance dose is below the EL shall be classified as belonging to the Exempt Group.
- For repetitively pulsed lamps, a lamp whose weighted radiant exposure or weighted radiance dose is below the EL, shall be evaluated using the Continuous wave risk criteria discussed in clause 6.1, using time averaged values of the pulsed emission.

Table 6.1 Emission limits for risk groups of continuous wave lamps.

Risk	Action spectrum	Symbol	Emission limits			Units
			Exempt	Low risk	Mod risk	
Actinic UV	$S_{UV}(\lambda)$	E_s	0,001	0,003	0,03	$W \cdot m^{-2}$
Near UV		E_{UVA}	10	33	100	$W \cdot m^{-2}$
Blue light	$B(\lambda)$	L_B	100	10000	4000000	$W \cdot m^{-2} \cdot sr^{-1}$
Blue light, small source	$B(\lambda)$	E_B	1,0*	1,0	400	$W \cdot m^{-2}$
Retinal thermal	$R(\lambda)$	L_R	$28000/\alpha$	$28000/\alpha$	$71000/\alpha$	$W \cdot m^{-2} \cdot sr^{-1}$
Retinal thermal, weak visual stimulus**	$R(\lambda)$	L_{IR}	$6000/\alpha$	$6000/\alpha$	$6000/\alpha$	$W \cdot m^{-2} \cdot sr^{-1}$
IR radiation, eye		E_{IR}	100	570	3200	$W \cdot m^{-2}$
* Small source defined as one with $\alpha < 0,011$ radian. Averaging field of view at 10000 s is 0,1 radian.						
** Involves evaluation of non-GLS source						

ANNEX C

(informative)

UNCERTAINTY ANALYSIS

The analysis of uncertainty requires that all sources of uncertainty be quantitatively evaluated. The first step in uncertainty analysis is identifying the various uncertainty sources. Below is a list that can be the start of the evaluation of uncertainty for measurements used in classifying lamps and lamp systems. See further details in CIE 53 – 1982 Methods of characterizing the performance of radiometers and

- photometers.
- Instrument
- Wavelength
- Bandwidth
- Response to varying spectra
- Linearity
- Stability
- Stray light
- Out-of-path
- Out-of-band
- Calibration
- Standards (assigned uncertainty)
- Distance
- Alignment
- Electrical operation
- Measurement
- Alignment
- Distance
- Source under test
- Stability
- Temperature
- Size
- Electrical operation

The individual uncertainty factors must be found or estimated. Each factor must then be propagated through the measurement to find the impact on the weighted values used in hazard classification. As seen in clause 5.3.3 above, the percent impact on the weighted values may be different from the percent uncertainty of the individual factor. This is seen in the change in output from an incandescent lamp as a function of the input current. The change in current changes the energy dissipated in the lamp and therefore changes the lamp's resistance and temperature. A change in current will not be followed linearly by a change in lamp's total output. Because of the change in temperature, the output at a single wavelength may not follow the change in total output. Though the changes are not linear, for small changes they can be considered linear. The normal way to propagate the uncertainty is to find the amount of the final

value's change to the initial factors change. This ratio of the impact to the input change is called sensitivity. For the incandescent lamp example, a change in the lamp's current by a small percent leads to a change in the output at the wavelength of interest as shown.

Table C.1 Example of uncertainty propagation.

	Lamp Current	Signal at 300 nm
Setting 1	8,2000	8451
Setting 2	8,2011	8461
Change	0,0134%	0,118%

Sensitivity is $0,118\% / 0,0134\% = 9$.

An uncertainty of 0,1% in current will be an uncertainty of 0,9% in output at 300 nm.

Each of the uncertainties must be carried through to the final value and expressed as a percentage.

The full sets of uncertainties are combined in quadrature and this combined uncertainty expressed as a percentage is then reported with the measured value.